Haptic Arm Navigation Aid

Electronics design, schematics and operating principle

1. What the device does

This is a wearable aid for blind and low-vision users. It tells the wearer about the space around them through **touch** instead of sound, so their hearing stays free. Two things are sensed and both are delivered as vibration on the forearm:

- **Things approaching or nearby** (people, vehicles, obstacles) in four directions: front, back, left, right.
- **Loss of ground**: a step down, kerb or drop-off, detected by how far the arm is from the floor.

The system is split into two parts joined by one thin cable: a compact **arm box** worn under the forearm (brain, battery, vibration motors, ground sensor, orientation sensor) and a flexible removable **neck collar** that carries four ultrasonic range sensors.

Architecture at a glance

- **Sense:** 4 ultrasonic sensors (collar), 1 IR distance sensor (arm box), 1 orientation sensor (arm box).
- **Think:** one ESP32 microcontroller reads everything, works out direction, distance, approach speed and arm tilt.
- **Act:** four coin vibration motors on the box (N, E, S, W). The motor that buzzes is the direction of the event; the pulse pattern says what kind of event it is and how urgent.

2. Physical layout

The arm box is a small, slim rectangle. The four motors sit at the midpoint of each side. The IR sensor sits inside the north end, just behind the north motor, facing the ground. The collar is a flexible band that springs back around the neck and can be unwrapped to remove it; its four sensors face front, right, back and left. North on the box always means the wearer's front.

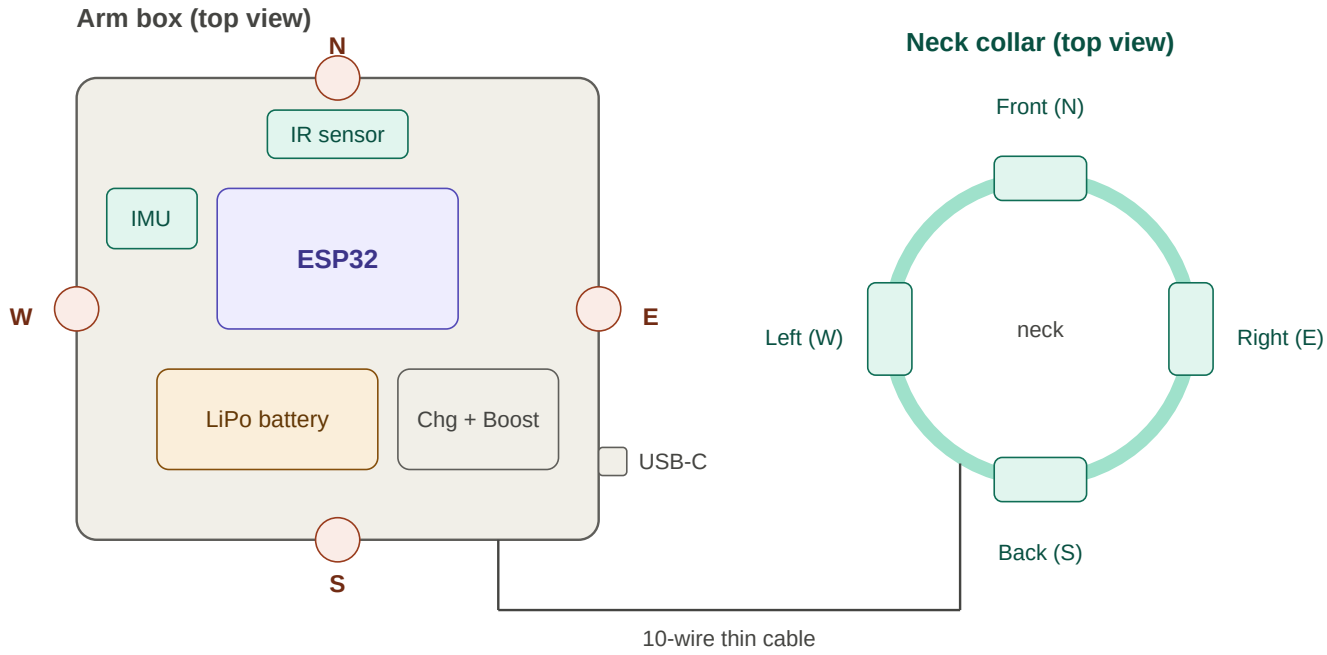

Figure 1. Physical arrangement of the arm box and the neck collar.

3. Parts list and what each part does

Part	Qty	Role in the system
ESP32 DevKit (30 or 38 pin)	1	The brain. Reads sensors, decides what to vibrate, runs on 3.3 V logic. Has enough pins, ADC and PWM channels for everything here.
BNO085 IMU breakout	1 (+1 optional)	Gives arm orientation (tilt and heading). Used so lifting or swinging the arm does not cause false "no ground" alarms, and to keep direction alerts correct as the arm turns. The optional second unit goes on the collar.
Sharp GP2Y0A02YK0F IR distance sensor	1	IR triangulation sensor, 20 to 150 cm, analog output. Measures arm-to-ground distance. Mounted in the north end of the box, facing down.
HC-SR04P ultrasonic sensor (3.3 V type)	4	Range finders on the collar: front, right, back, left. Time-of-flight echo gives distance; change in distance over time gives approach speed.
Coin ERM vibration motor, 3 V, about 10 mm	4	The output. One on each side of the box (N, E, S, W).
AO3400A N-MOSFET (or similar logic-level)	4	Electronic switch for each motor. The ESP32 pin cannot supply motor current directly.
1N5819 Schottky diode	4	Flyback diode across each motor, absorbs the voltage spike when the motor switches off.
Resistors: 100 ohm x4, 100 kohm x6	10	100 ohm limits gate current. 100 kohm pulldowns keep motors off while the ESP32 boots (4), plus 2 for the battery voltage divider.
LiPo battery, 1S 3.7 V, about 1000 mAh, with protection	1	Compact flat pack (RC-style). Runs everything.
TP4056 charger module with USB-C and protection	1	Charges the LiPo from a USB-C cable and guards against over-discharge and short circuit.
MT3608 boost converter module	1	Raises battery voltage to a steady 5.0 V for the ESP32 VIN pin and the Sharp sensor.
Slide switch	1	Main power on/off between battery and the rest of the circuit.
Capacitors: 100 uF, 10 uF, 0.1 uF	few	100 uF on the 5 V rail near the ESP32; 10 uF right at the Sharp sensor supply (it draws current spikes); 0.1 uF decoupling on sensors.
10-pin connector and thin flexible cable	1 set	Detachable link between arm box and collar. Small JST-SH or magnetic pogo connector so the collar can be removed.
Flexible collar band	1	Spring-steel or TPU strip in a soft sleeve that wraps the neck when released and unwraps by hand.

4. Schematics

4.1 Main wiring (ESP32 pin map)

Every sensor and driver connects to the ESP32 as shown. Echo lines use input-only pins (34, 36, 39) where possible, trigger and motor lines use normal output pins, and boot-strapping pins are avoided so the board starts reliably. The Sharp sensor is on an ADC1 pin so it keeps working if radio features are used later.

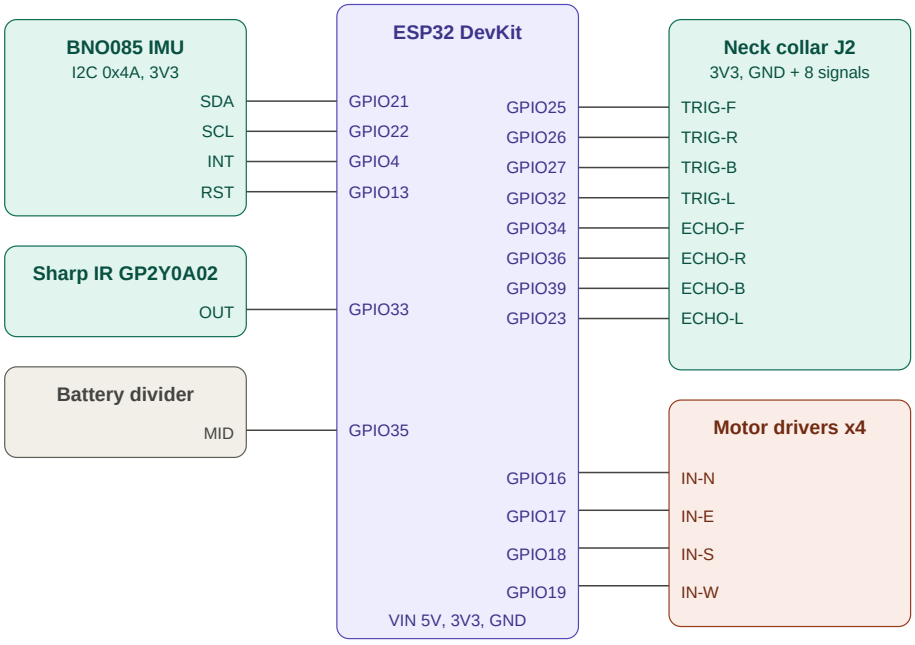

Figure 2. Signal wiring between the ESP32 and all modules. 3V3 and GND go to every sensor (not drawn).

4.2 Collar cable pinout (J2, 10 conductors)

Pin	Signal	Goes to	Pin	Signal	Goes to
1	3V3	All 4 sensors VCC	6	TRIG-L	GPIO32
2	GND	All 4 sensors GND	7	ECHO-F	GPIO34
3	TRIG-F	GPIO25	8	ECHO-R	GPIO36
4	TRIG-R	GPIO26	9	ECHO-B	GPIO39
5	TRIG-B	GPIO27	10	ECHO-L	GPIO23

If a second BNO085 (address 0x4B) is added on the collar, add SDA and SCL to the cable for a 12-wire link.

4.3 Power chain

The battery powers a 5 V boost converter, which feeds the ESP32 VIN pin and the Sharp sensor. The ESP32's own 3.3 V regulator feeds the BNO085 and the four ultrasonic sensors. The vibration motors are connected to the battery rail directly (not the 3.3 V rail) so their current does not overload the ESP32 regulator. A 100 kohm / 100 kohm divider lets the ESP32 read battery level.

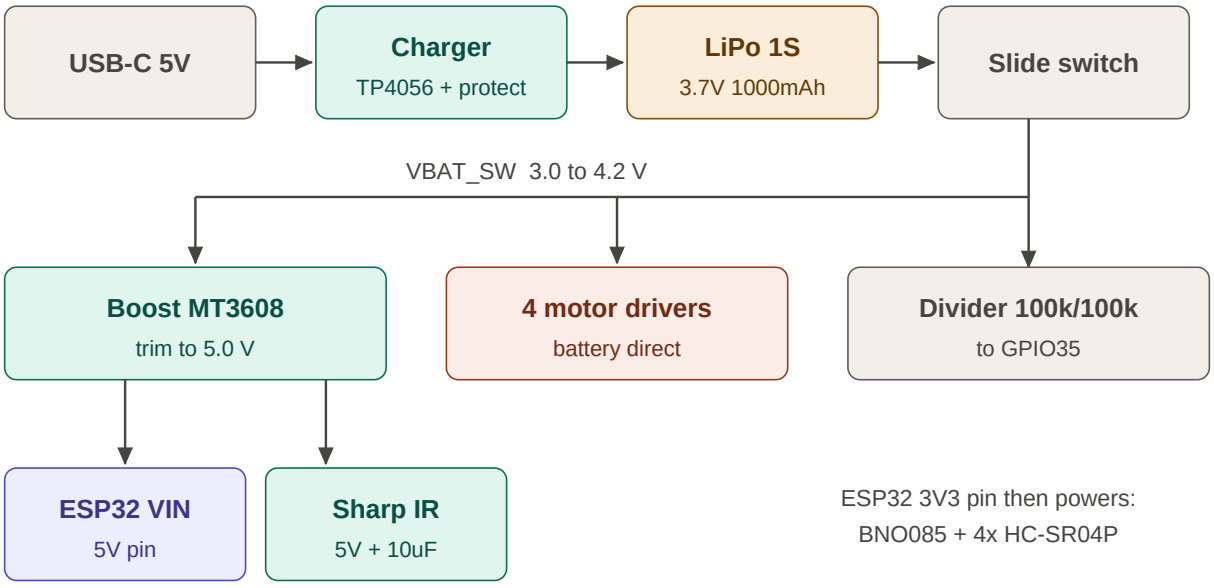

Figure 3. Power distribution.

4.4 Motor driver (one channel, repeated for N, E, S, W)

The ESP32 pin drives the MOSFET gate through 100 ohm. A 100 kohm resistor to ground keeps the gate low during boot so no motor buzzes randomly. The motor sits between the battery rail and the MOSFET drain; the flyback diode across it protects the MOSFET. PWM controls strength. Limit PWM duty to about 70 percent so the average voltage stays near the motor's 3 V rating even with a full 4.2 V cell.

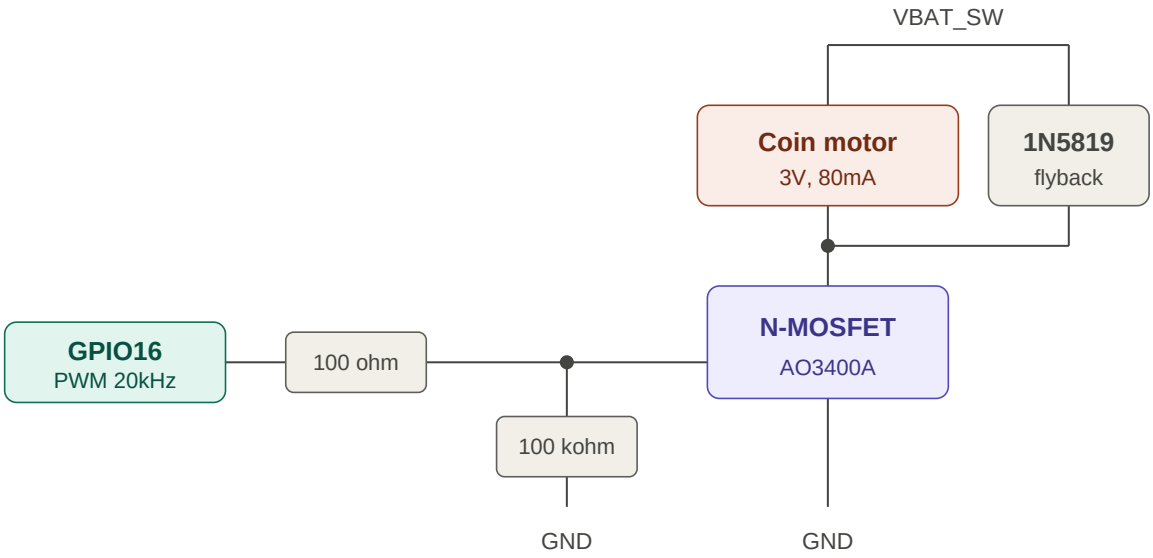

Figure 4. Single motor driver channel. Channels use GPIO16 (N), GPIO17 (E), GPIO18 (S), GPIO19 (W).

5. How it works

5.1 Obstacle and approach sensing

The ESP32 fires the four ultrasonic sensors one at a time (about 30 ms apart) so one sensor never hears another's echo. One full sweep takes roughly 120 ms, about 8 updates per second. Distance is time-of-flight times the speed of sound divided by two. Speed is the change in distance between sweeps divided by the time between them, smoothed over a few readings.

5.2 Direction to motor mapping

- Front sensor triggers the North motor, right sensor the East motor, back sensor the South motor, left sensor the West motor.
- Diagonal: when two adjacent sensors both see the object, both of their motors vibrate together (North + West = front-left).
- Beam note: each HC-SR04P has a narrow beam (roughly 15 degrees), so with four sensors at 90 degrees there are blind gaps at the diagonals. Wider-beam sensors or slightly angled mounting reduce this.

5.3 Vibration language

Coin motors change strength with PWM but not true frequency, so different events are told apart by pulse rhythm and strength, plus which motors are involved.

Event	Motors	Pattern	Strength
Slow-walking person or object approaching	Direction motor	Slow single pulses (about 1 per second)	Medium, rises as distance falls
Running person / fast approach	Direction motor	Rapid short pulses (about 6 per second)	Strong
Vehicle / very fast object	Direction motor	Long-long-short repeating, no gaps	Maximum
Static obstacle ahead	Direction motor	Soft slow pulses	Low to medium by distance
Ground lost (drop-off, step down)	All four together	Continuous buzz, clearly unlike any directional pattern	Scales with size of the drop
Low battery	All four	Two short buzzes at start-up and periodically	Low

5.4 Ground-loss (drop-off) detection

At setup, the wearer stands normally with the arm relaxed and the device records the average arm-to-ground distance; this is calibrated once in production and can be re-calibrated with a button. During use the Sharp sensor reading is compared with that baseline. If the distance grows beyond a tolerance, the drop alert fires, and the amount of extra distance sets how strong the vibration is.

To stop false alarms, the BNO085 tells the ESP32 how the forearm is tilted. The ground reading is trusted only when the forearm is near its calibrated rest tilt, and it can be corrected by the cosine of the tilt angle. When the arm is raised or swung, ground readings are ignored for that moment.

5.5 Keeping directions correct as the arm moves

The collar sensors are fixed to the body, but the arm box turns with the forearm. If the arm swings out to the side, the box's north is no longer the wearer's front. The fix is to rotate each alert by the arm's heading relative to the body. With one BNO085 this works after a "stand straight" calibration while the torso is not turning. A second BNO085 on the collar gives the body heading directly and makes this robust.

5.6 Firmware loop (outline)

- Read BNO085 (tilt and heading) and battery level.
- Ping each ultrasonic sensor in turn; store distance and timestamp.
- Work out approach speed for each side; classify event (walk, run, vehicle, static).
- Read the Sharp sensor; if arm tilt is valid, compare with baseline and compute drop size.
- Rotate directions by arm heading; choose motors (single or adjacent pair).
- If a drop is active it takes priority and drives all four motors; otherwise play the directional pattern.
- Set PWM (capped at about 70 percent) and repeat.

6. Build and test notes

- Set the MT3608 to exactly 5.0 V with a multimeter before connecting the ESP32.
- Use HC-SR04P (3.3 V) parts. A standard 5 V HC-SR04 needs a 1k / 2k divider on each ECHO line.
- Keep motor wiring short and away from the ultrasonic and IR sensor wiring to reduce noise.
- Mount the Sharp sensor facing straight down with a clear window; strong sunlight can disturb IR sensors, so test outdoors.
- Expected runtime: roughly 3 to 5 hours on a 1000 mAh cell; measure with real use and size the battery to match.
- Test with a sighted helper in a safe space first. This device supplements, and does not replace, a cane or guide dog.

Appendix: ESP32 pin summary

Function	GPIO	Function	GPIO
I2C SDA / SCL	21 / 22	TRIG F / R / B / L	25 / 26 / 27 / 32
BNO085 INT / RST	4 / 13	ECHO F / R / B / L	34 / 36 / 39 / 23
Sharp IR output (ADC1)	33	Motor N / E / S / W	16 / 17 / 18 / 19
Battery divider (ADC1)	35	Power in	VIN = 5 V, 3V3 out, GND